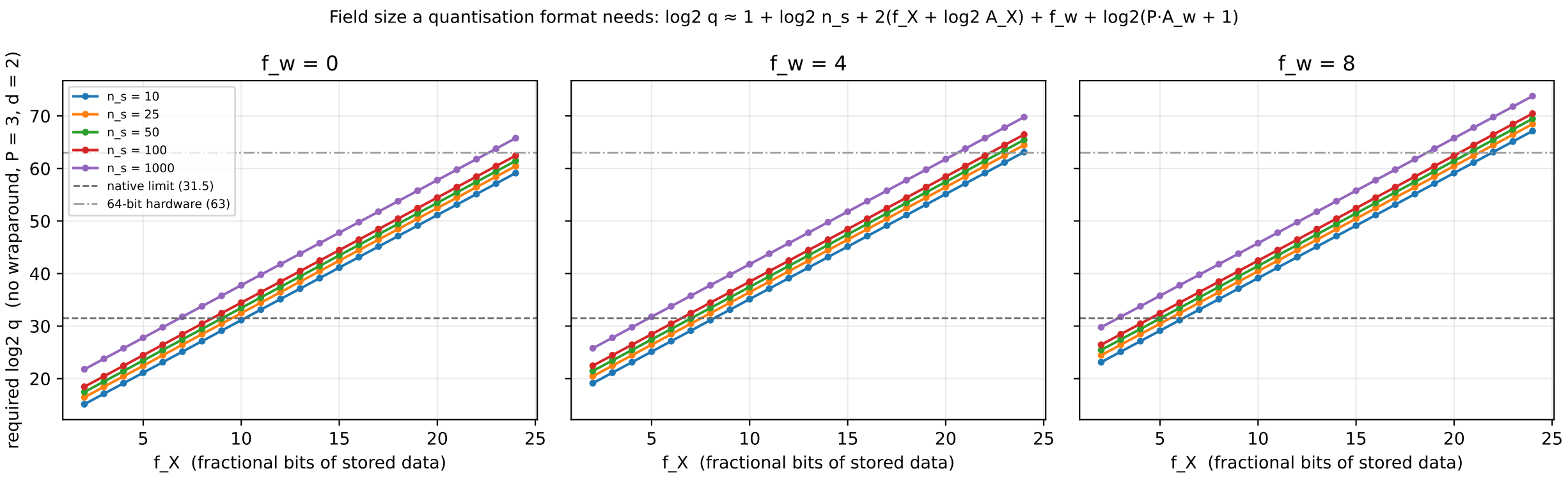
Field size a quantisation format needs: log2 q ≈ 1 + log2 n_s + 2(f_X + log2 A_X) + f_w + log2(P·A_w + 1)
f_w = 0
f_w = 4
f_w = 8
n_s = 10
n_s = 25
n_s = 50
n_s = 100
n_s = 1000
native limit (31.5)
64-bit hardware (63)
required log2 q (no wraparound, P = 3, d = 2)
f_X (fractional bits of stored data)
20
30
40
50
60
70
5
10
15
20
25